Firewall, NAT and Packet Filtering Assignment

Note: Actual Windows Firewall screenshot must be captured on your own PC and inserted where indicated.

1. Windows Defender Firewall - Block Port 8080

• Open Windows Defender Firewall with Advanced Security.

• Select Inbound Rules -> New Rule.

• Choose Port -> TCP -> Specific local ports: 8080.

• Select Block the connection.

• Apply to Domain, Private and Public profiles.

• Name the rule: Block_TCP_8080.

SCREENSHOT PLACEHOLDER: Insert screenshot of the created inbound rule here.

2. NAT Simulation Diagram

[Phone 192.168.1.10]\
[Laptop 192.168.1.20] ----> [WiFi Router NAT] ----> [Public IP 49.36.120.50] ----> Internet
[Smart TV 192.168.1.30]/

The router translates multiple private IP addresses to one public IP address before traffic reaches the Internet.

3. Port Filtering - Block Outgoing Port 25 (SMTP)

Router Steps: Login to router admin panel -> Security/Access Control -> Port Filtering -> Add rule -> Outbound -> Block TCP Port 25 -> Save.

Blocking outbound SMTP traffic on port 25 can help stop infected devices from sending spam emails directly. It reduces the risk of your network being abused for unsolicited email campaigns.

4. Packet Filtering Analysis

The trainer's packet filtering rule set was not provided. Therefore the packets cannot be definitively classified as Allowed or Blocked. Evaluate each packet against the provided source IP, destination IP, protocol and port rules.

Packet	Status
HTTP 192.168.1.5 -> 172.217.166.78:80	Depends on rule set
SSH 10.0.0.10 -> 34.201.23.1:22	Depends on rule set
DNS 192.168.1.7 -> 8.8.8.8:53	Depends on rule set

5. How Instagram or Zomato May Use Firewalls and Packet Filtering

• Block malicious traffic such as port scans, unauthorized access attempts and known attack signatures.

• Use Web Application Firewalls (WAFs) to filter suspicious HTTP requests and protect APIs and web applications.

• Rate-limit or filter packets from abusive sources to reduce bot activity, denial-of-service attacks and automated misuse.